

SOWMYA KALIMERA

Jersey City, NJ | +1 5515548927 | sowmyakalimera@gmail.com | [Linkedin](#)

EDUCATION

Pace University, Seidenberg School of Computer Science and Information Systems

New York, NY

Master of Science in Information Systems (GPA: 3.10/4.0)

Sep 2023 - May 2025

OSMANIA UNIVERSITY

Hyderabad, India

Bachelor of Science in Computer Science (GPA: 3.88/4.0)

Jul 2017 - Oct 2020

WORK EXPERIENCE

Capgemini Technology Services India Ltd.

Software Engineer, Hyderabad, India

Nov 2021 - Dec 2023

- Developed and maintained 20+ scalable microservices using Java, Spring Boot, Hibernate, and JPA to process financial transactions; reduced API response time by 35% and improved throughput for 3M+ monthly trades across payment and credit systems.
- Built secure RESTful APIs with OAuth2, Spring Security, and JWT; enforced fine-grained role-based access control (RBAC) for internal and external users, ensured PCI-DSS and SOX compliance, and prevented 80% of previously untracked unauthorized access attempts.
- Automated CI/CD pipelines using Jenkins, GitHub Actions, Maven, and Nexus; introduced SonarQube quality gates, environment-based deploy logic, and rollback triggers—cut release cycle from 5 to 2 days and reduced production rollback errors by 50%.
- Tuned PostgreSQL and Oracle batch jobs for daily settlement, reconciliation, and reporting; introduced partitioned queries, indexing, and batching logic to reduce job runtime by 40% and support time-sensitive analytics across 10+ downstream teams.
- Integrated Prometheus, Grafana, and custom JVM metrics into 30+ microservices; enabled anomaly detection and live dashboards for memory usage, response latency, and service health—reduced MTTR by 50% and sustained 99.9% uptime in production.
- Participated in Agile sprints across cross-functional teams; led sprint planning, task estimations, peer code reviews, and grooming sessions—boosted delivery velocity by 30%, reduced rework by 50%, and ensured sprint predictability.
- Achieved 85% test coverage using JUnit, Mockito, and TestContainers; integrated regression and unit tests into Maven CI workflows, enabling early-stage bug detection, reducing QA turnaround by 60%, and lowering production defect rate by 70%.

Accolite Digital

Software Development Intern, Hyderabad, India

Jun 2021 - Oct 2021

- Built Spring Boot APIs for healthcare analytics, integrating Hibernate and PostgreSQL for efficient data access; optimized SQL joins and caching strategies, reducing reporting latency by 35% and improving response times for high-frequency requests.
- Developed responsive, real-time dashboards using Angular and RxJS to visualize patient vitals, appointment schedules, and alerts; modularized components and reduced UI load time by 40%, enhancing daily workflow for over 100 clinical staff.
- Achieved 90% test coverage across backend and frontend layers using Postman, JUnit, Mockito, and GitHub Actions CI workflows; accelerated defect discovery during QA cycles and improved test automation reliability across environments.
- Participated in Agile ceremonies with globally distributed teams using JIRA and Confluence; contributed to user story refinement, effort estimations, and sprint retrospectives, improving sprint velocity by 25% and reducing rework on assigned features.

SKILLS

Languages & Frameworks: Java, Python, JavaScript, C#, Bash, Spring Boot, .NET, React.js, Angular, Node.js, Hibernate, JPA, Servlets

Databases & ORM/Testing: PostgreSQL, MySQL, MongoDB, Oracle, Hibernate, JPA, SQL, JUnit, Mockito, Selenium, Cypress, Postman,

Cloud & DevOps: AWS (EC2, RDS, Lambda), Azure, GCP, Docker, Kubernetes, Jenkins, Terraform, GitHub Actions, GitLab CI/CD

Monitoring & Concepts: Prometheus, Grafana, Splunk, ELK Stack, WebSockets, Git, GitHub, GitLab, Jira, Confluence, REST APIs, OAuth2, JWT, RBAC, Microservices, DevSecOps, CI/CD, Secure SDLC, Agile/Scrum, OOP

PROJECTS

Student Learning Analytics Platform

- Built a full-stack academic dashboard for 5K+ students, enabling GPA trends, course outcomes, and attendance-based risk flags; used JWT, RBAC, and role-based views to securely restrict and personalize data access for students, faculty, and admins.
- Reduced dashboard latency by 60% by optimizing SQL aggregations, materialized views, and advanced indexing in PostgreSQL; enabled early intervention alerts for faculty through automated KPI insights, threshold detection, and email triggers.
- Containerized backend and frontend services with Docker; deployed to AWS using GitHub Actions for CI/CD automation, enabling reproducible builds, reliable version control, and faster feature releases with <10 min pipeline turnaround.

Credit Risk Analytics Engine

- Built a full-stack credit risk engine simulating bond ratings and CDS alerts; processed over 1M synthetic trades using Apache Kafka pipelines and stored risk outputs in MongoDB, enabling real-time dashboards and historical analytics.
- Designed a responsive React.js UI showing credit scores, watchlists, and trade signals; reduced analyst lookup time by 45%, improved model clarity, and supported interactive simulations for 50+ market scenarios with sub-second updates.
- Deployed services to AWS EC2 using Jenkins and GitHub Actions; implemented hybrid batch-streaming ingestion, cutting risk evaluation latency by 65% and scaling throughput to handle 500K+ trade evaluations/hour during stress tests.